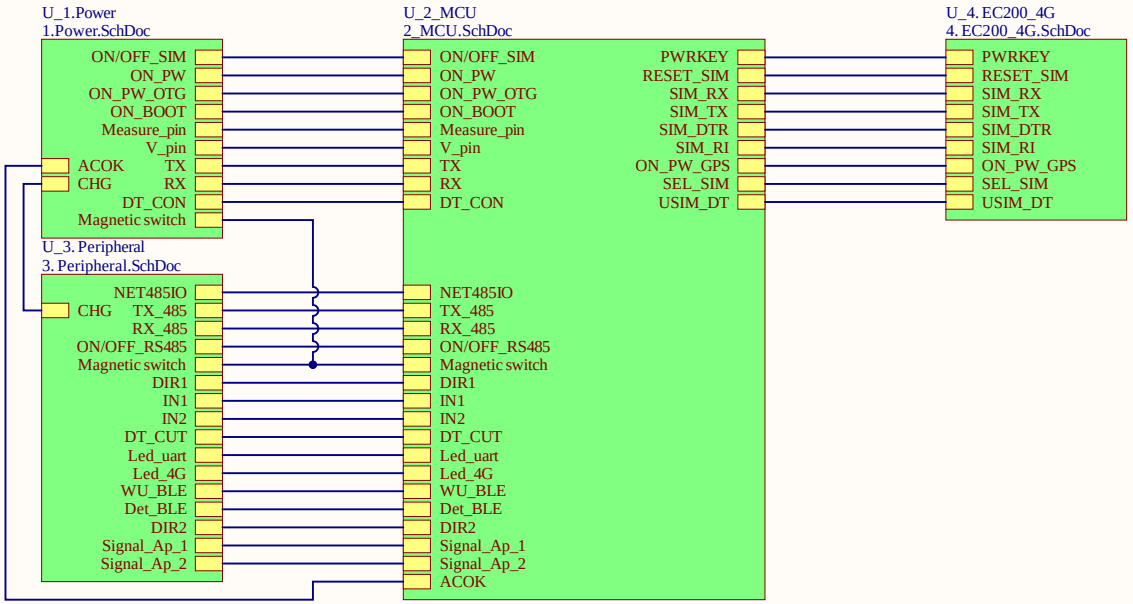
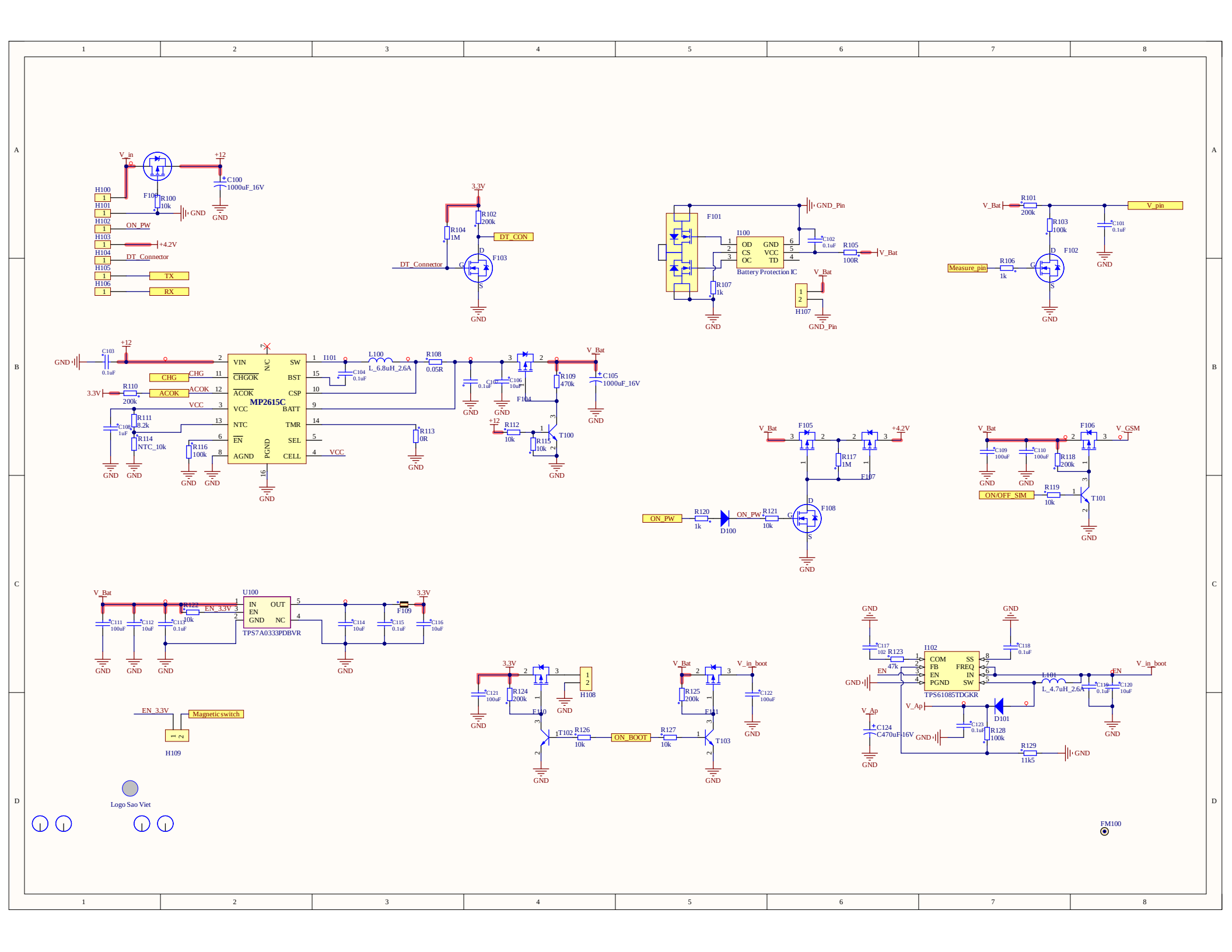
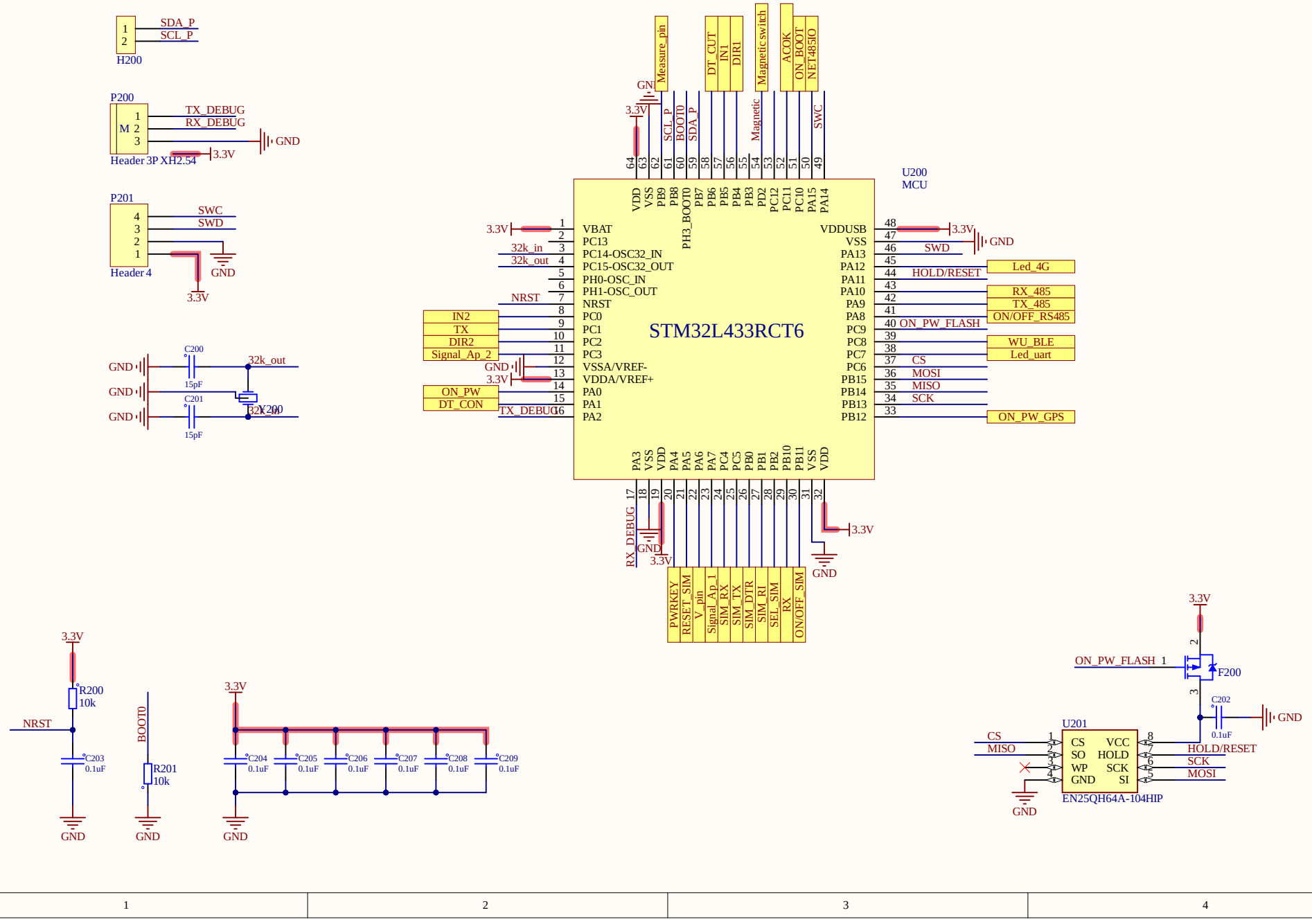
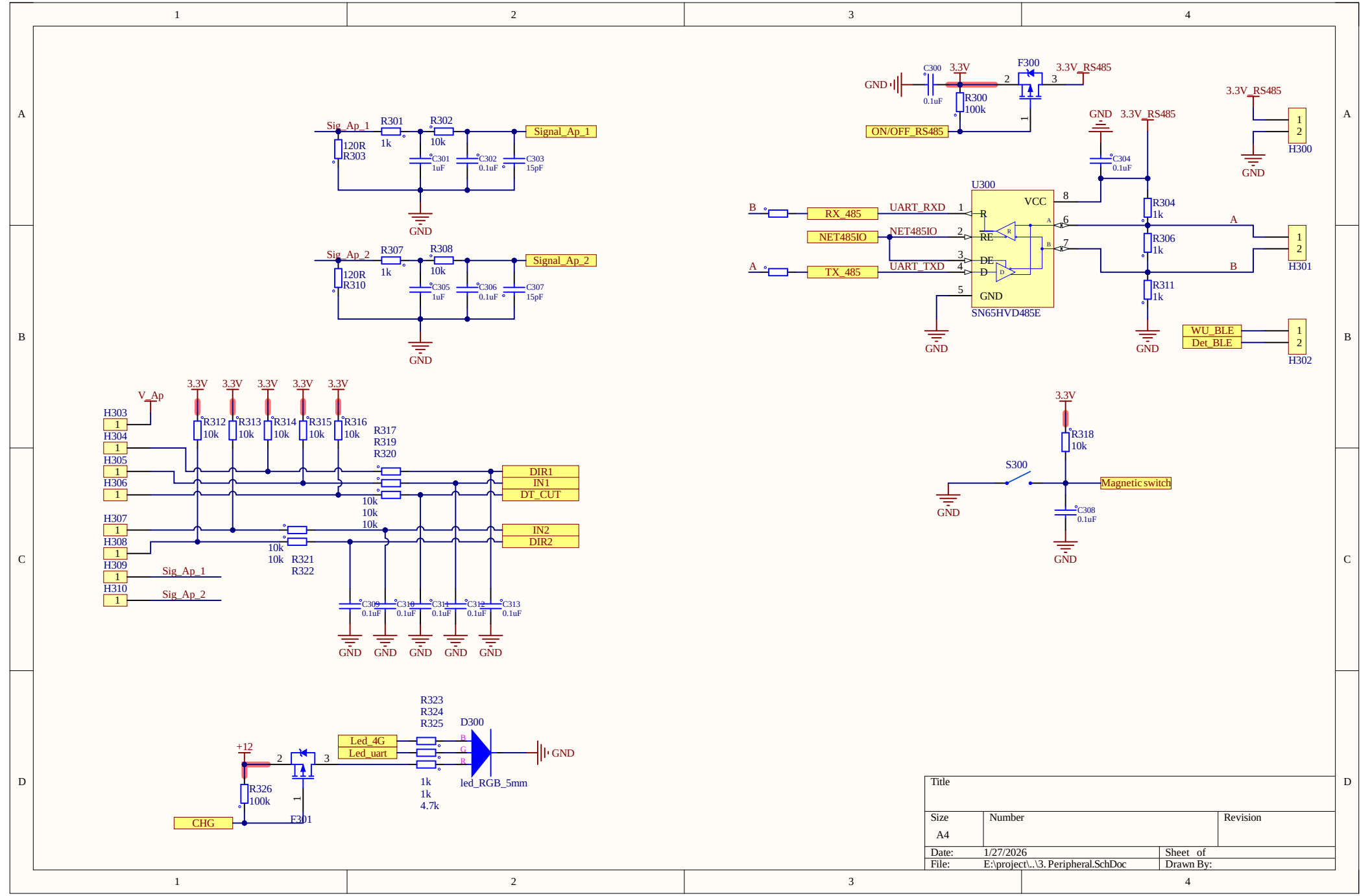
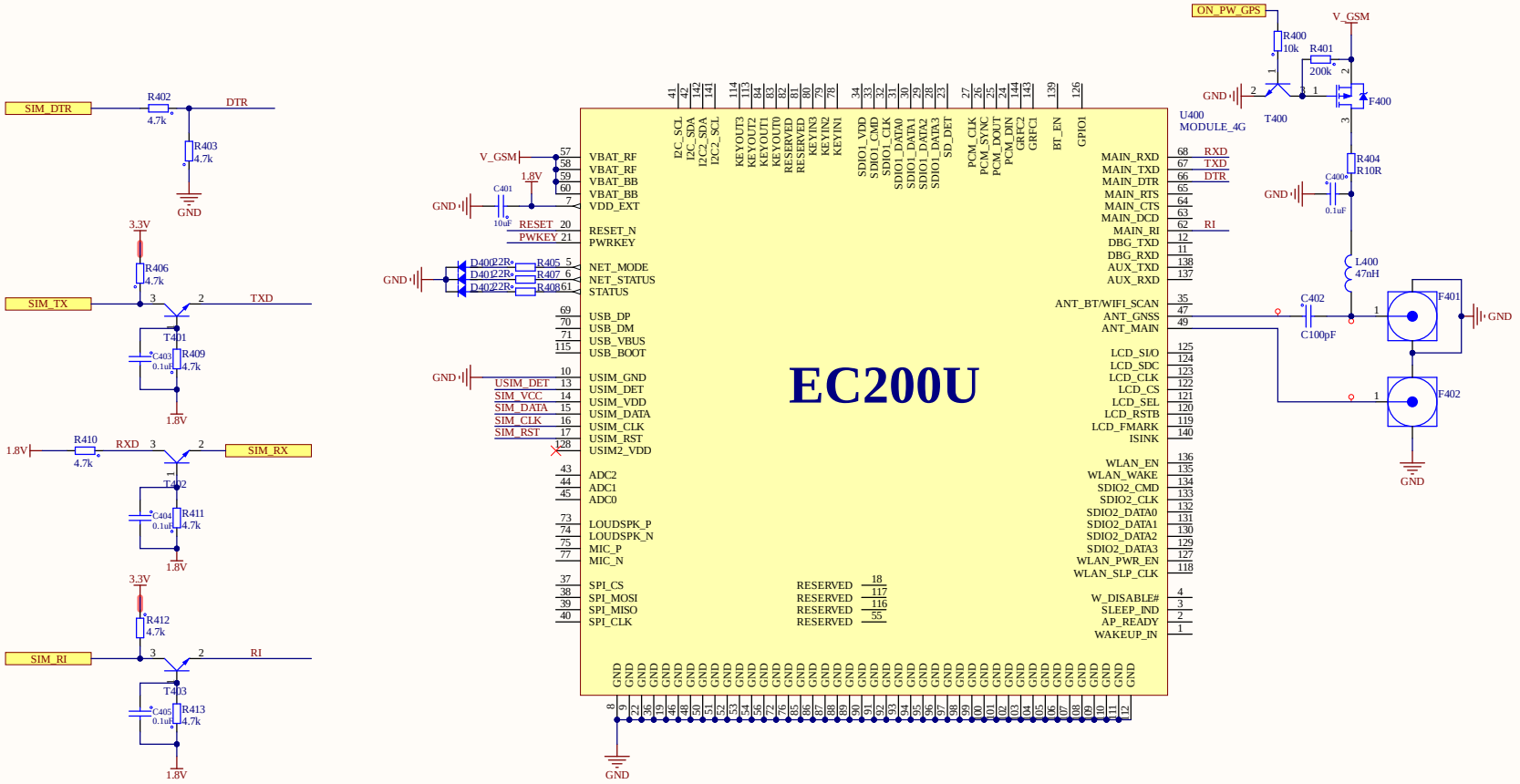


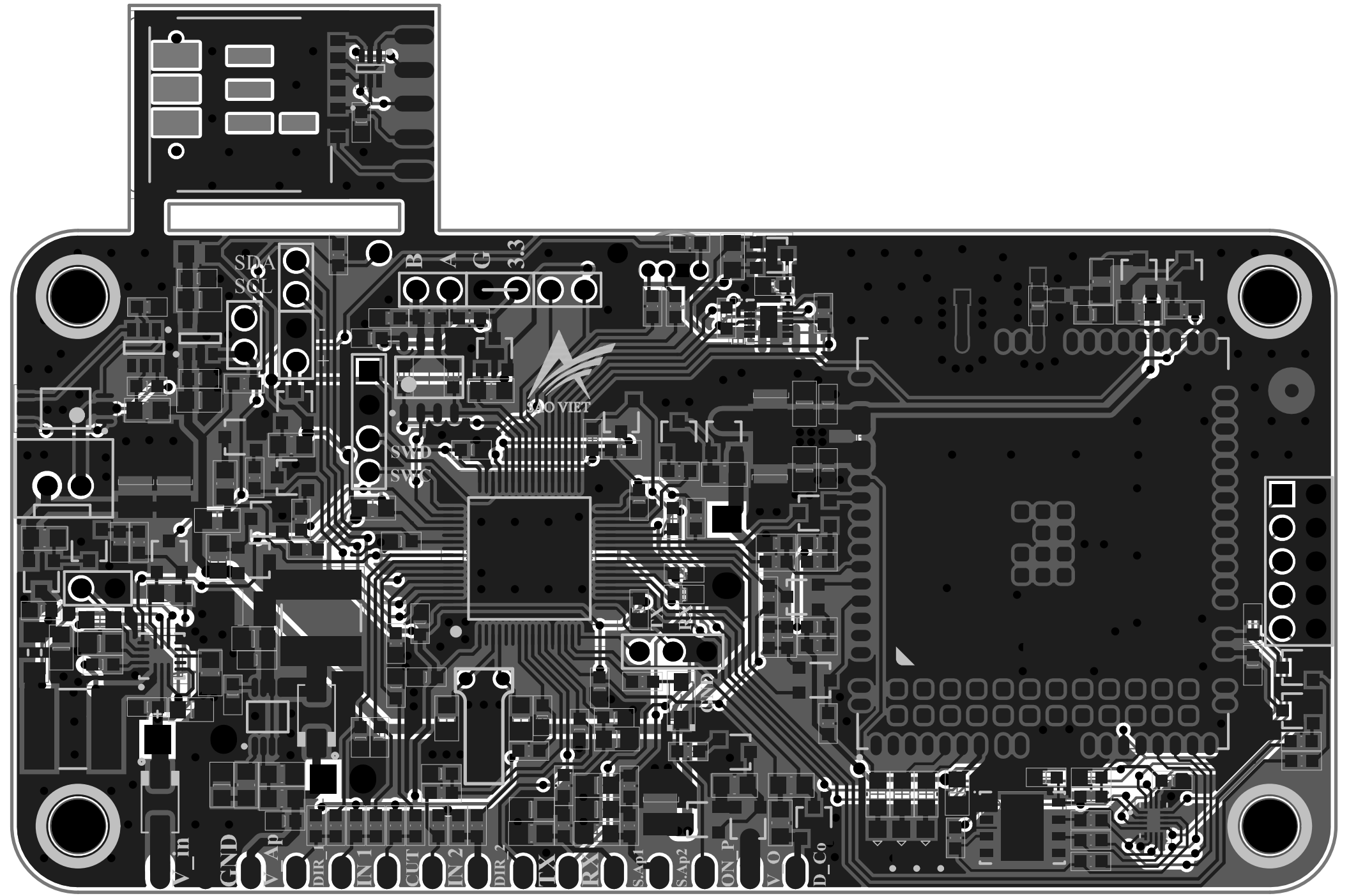
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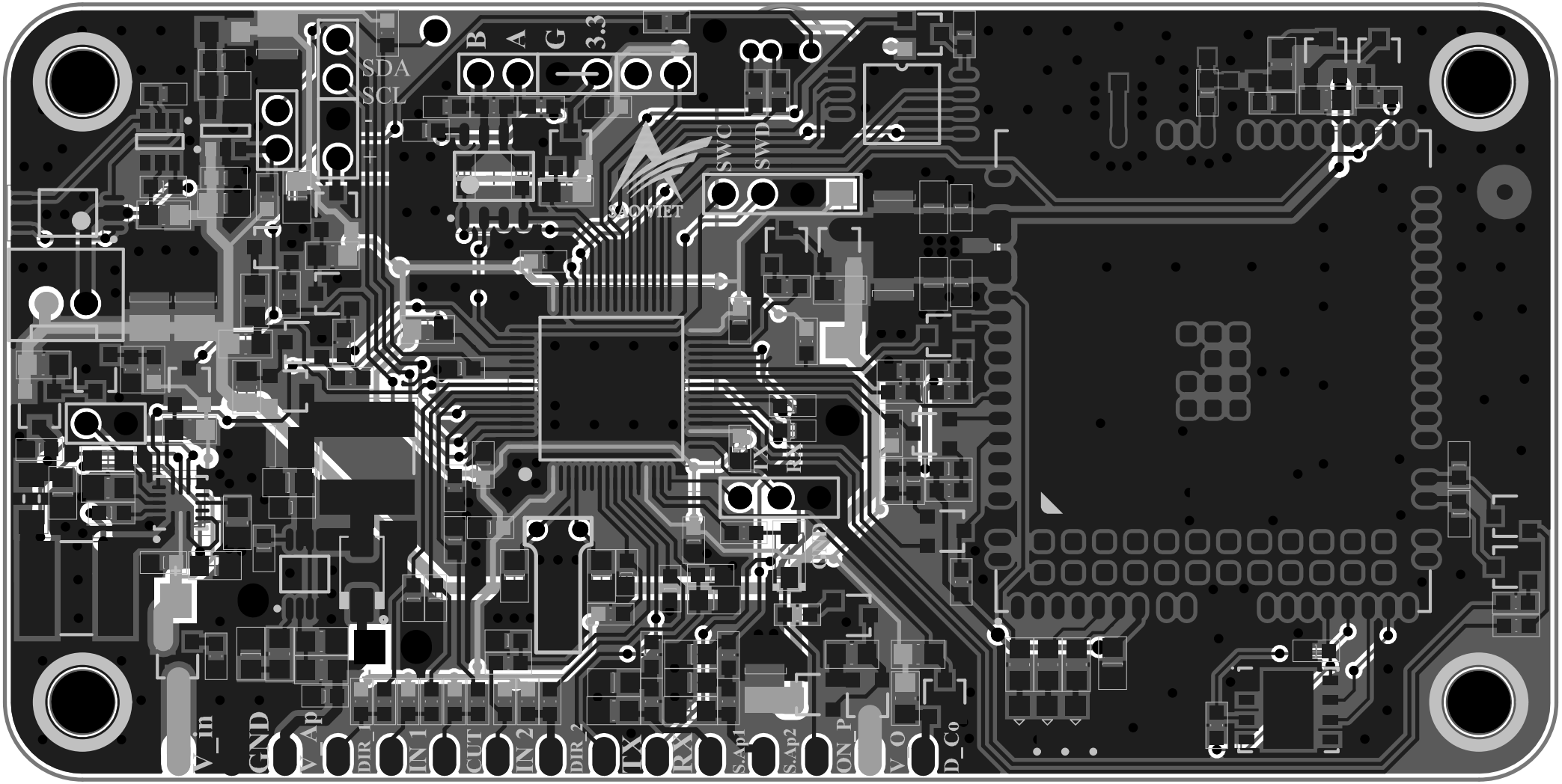

V_in
+12
C100
1000uF_16V
H100
F100
R100
10k
GND
H101
H102
ON_PW
H103
+4.2V
H104
DT_Connector
H105
TX
H106
RX
3.3V
R102
200k
R104
1M
DT_CON
DT_Connector
F103
F101
GND_Pin
I100
OD
GND
CS
VCC
OC
TD
Battery Protection IC
C102
0.1uF
R105
100R
V_Bat
R107
1k
H107
V_pin
R101
200k
R103
100k
C101
0.1uF
F102
Measure_pin
R106
1k
C103
0.1uF
CHG
ACOK
VCC
MP2615C
VIN
CHGOK
ACOK
NTC
AGND
N/C
SW
BST
CSP
BATT
TMR
SEL
CELL
PGND
I101
L100
L_6.8uH_2.6A
R108
0.05R
C104
0.1uF
C105
1000uF_16V
R109
470k
F104
R110
200k
R111
8.2k
R114
NTC_10k
R116
100k
R113
0R
R112
10k
R115
10k
T100
C106
10uF
C107
0.1uF
F105
F106
F107
F108
R117
1M
R118
200k
R119
10k
T101
V_GSM
ON/OFF_SIM
R120
1k
D100
R121
10k
+4.2V
U100
TPS7A0333PDBVR
IN
OUT
EN
GND
NC
R122
10k
EN_3.3V
F109
C111
100uF
C112
10uF
C113
0.1uF
C114
10uF
C115
0.1uF
C116
10uF
H108
R124
200k
R125
200k
T102
R126
10k
ON_BOOT
R127
10k
T103
V_in_boot
C121
100uF
C122
100uF
Magnetic switch
H109
Logo Sao Viet
I102
COM
FB
EN
PGND
SS
FREQ
IN
SW
TPS61085TDGKR
R123
47k
C117
102
C118
0.1uF
L101
L_4.7uH_2.6A
C119
0.1uF
C120
10uF
V_Ap
C124
C470uF-16V
C123
0.1uF
R128
100k
D101
R129
11k5
FM100


H200
SDA_P
SCL_P
P200
TX_DEBUG
RX_DEBUG
GND
3.3V
Header 3P XH2.54
P201
SWC
SWD
Header 4
C200
C201
15pF
32k_out
32k_in
U200
MCU
STM32L433RCT6
VBAT
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PH0-OSC_IN
PH1-OSC_OUT
NRST
PC0
PC1
PC2
PC3
VSSA/VREF-
VDDA/VREF+
PA0
PA1
PA2
VDDUSB
VSS
PA13
PA12
PA11
PA10
PA9
PA8
PC9
PC8
PC7
PC6
PB15
PB14
PB13
PB12
PH3_BOOT0
IN2
TX
DIR2
Signal_Ap_2
ON_PW
DT_CON
TX_DEBUG
Measure_pin
DT_CUT
IN1
DIR1
Magnetic switch
ACOK
ON_BOOT
NET485IO
Magnetic
SWC
SCL_P
BOOT0
SDA_P
Led_4G
HOLD/RESET
RX_485
TX_485
ON/OFF_RS485
ON_PW_FLASH
WU_BLE
Led_uart
CS
MOSI
MISO
SCK
ON_PW_GPS
PWRKEY
RESET_SIM
V_pin
Signal_Ap_1
SIM_RX
SIM_TX
SIM_DTR
SIM_RI
SEL_SIM
RX
ON/OFF_SIM
R200
10k
R201
10k
C203
C204
C205
C206
C207
C208
C209
0.1uF
F200
C202
U201
EN25QH64A-104HIP
CS
SO
WP
GND
VCC
HOLD
SCK
SI


Sig_Ap_1
R301
1k
R302
10k
120R
R303
C301
1uF
C302
0.1uF
C303
15pF
Signal_Ap_1
GND
Sig_Ap_2
R307
1k
R308
10k
120R
R310
C305
1uF
C306
0.1uF
C307
15pF
Signal_Ap_2
GND
GND
C300
0.1uF
3.3V
F300
3.3V_RS485
R300
100k
ON/OFF_RS485
GND
3.3V_RS485
C304
0.1uF
3.3V_RS485
H300
GND
U300
VCC
B
RX_485
UART_RXD
1
R
NET485IO
2
RE
A
TX_485
UART_TXD
3
DE
4
D
5
GND
SN65HVD485E
8
6
7
R304
1k
R306
1k
R311
1k
A
B
H301
GND
GND
WU_BLE
Det_BLE
H302
V_Ap
3.3V
3.3V
3.3V
3.3V
3.3V
H303
H304
H305
H306
H307
H308
H309
H310
R312
10k
R313
10k
R314
10k
R315
10k
R316
10k
R317
R319
R320
10k
10k
10k
DIR1
IN1
DT_CUT
IN2
DIR2
10k
10k
R321
R322
Sig_Ap_1
Sig_Ap_2
C309
0.1uF
C310
0.1uF
C311
0.1uF
C312
0.1uF
C313
0.1uF
GND
GND
GND
GND
GND
3.3V
R318
10k
S300
Magnetic switch
C308
0.1uF
GND
GND
+12
2
3
1
F301
R326
100k
CHG
R323
R324
R325
Led_4G
Led_uart
D300
B
G
R
1k
1k
4.7k
led_RGB_5mm
GND
Title
Size
A4
Number
Revision
Date:
1/27/2026
Sheet of
File:
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Drawn By:


EC200U
U400
MODULE_4G
VBAT_RF
VBAT_BB
VDD_EXT
RESET_N
PWRKEY
NET_MODE
NET_STATUS
STATUS
USB_DP
USB_DM
USB_VBUS
USB_BOOT
USIM_GND
USIM_DET
USIM_VDD
USIM_DATA
USIM_CLK
USIM_RST
USIM2_VDD
ADC2
ADC1
ADC0
LOUDSPK_P
LOUDSPK_N
MIC_P
MIC_N
SPI_CS
SPI_MOSI
SPI_MISO
SPI_CLK
RESERVED
MAIN_RXD
MAIN_TXD
MAIN_DTR
MAIN_RTS
MAIN_CTS
MAIN_DCD
MAIN_RI
DBG_TXD
DBG_RXD
AUX_TXD
AUX_RXD
ANT_BT/WIFI_SCAN
ANT_GNSS
ANT_MAIN
LCD_SI/O
LCD_SDC
LCD_CLK
LCD_CS
LCD_SEL
LCD_RSTB
LCD_FMARK
ISINK
WLAN_EN
WLAN_WAKE
SDIO2_CMD
SDIO2_CLK
SDIO2_DATA0
SDIO2_DATA1
SDIO2_DATA2
SDIO2_DATA3
WLAN_PWR_EN
WLAN_SLP_CLK
W_DISABLE#
SLEEP_IND
AP_READY
WAKEUP_IN
SIM_DTR
SIM_TX
SIM_RX
SIM_RI
V_GSM
1.8V
3.3V
GND
RESET
PWKEY
ON_PW_GPS
RXD
TXD
DTR
RI
T400
F400
F401
F402
L400
47nH
E400
MFF2
SIM_VCC
SIM_RST
SIM_CLK
SIM_DATA


SDA
SCL
B
A
C
3.3
SAO VIET
SWD
SWC
V_in
GND
V_Ap
DIR_
IN 1
CUT
IN 2
DIR_2
TX
RX
S.Ap1
S.Ap2
ON_P
V_O
D_Co


SDA
SCL
B
A
G
3.3
SAOVIET
SWC
SWD
TX
RX
V_in
GND
V_Ap
DIR_1
IN 1
CUT
IN 2
DIR_2
TX
RX
S.Ap1
S.Ap2
ON_P
V_O
D_Co

Comment	Description	Designator	Footprint	LibRef	Quantity	Manufacturer Part Number
4R	4R	R305, R309	603023	4R_R_0805	2	
0R	RES SMD 0 OHM 5% 1/10W 0603	R113	603025	R100k_0603	1	
0.05R	Res 25 mOhm 1% 1/2W 2256	R138	612306	R22R_1206	1	RE1206FR-0470R22L
0.1uF	Cap SMT Cer .1uF 50V ±5% 0603 AVTR, 0.1µF 16V Ceramic Capacitor X7R 0603 (1608 Metric) 0.063" L x 0.031" W (1.60mm x 0.80mm), 0.1µF ±10% 50V Ceramic Capacitor X7R 0805	C101, C102, C103, C104, C107, C113, C115, C118, C119, C123, C202, C203, C204, C205, C206, C207, C208, C209, C300, C302, C304, C306, C308, C309, C310, C311, C312, C313, C403, C403, C404, C405, C406, C408, C410, C411, C414	C0603	C0.1uF~0805, C0.1uF~0603	37	GMC103X7R104K50NT, EMK107B7104KA-T, C1751B104M5RCNA-NC
1k	RES SMD 1K OHM 5% 1/8W 0805	R138, R107, R120, R201, R204, R206, R207, R311, R323, R324	R0603	R1k_0805	10	CR0805-8W-1002T
1M	SMD 470K OHM 1% 1/8W 0805	R104, R117	603025	R200k_0805	2	
1uF	0.1µF 50V Ceramic Capacitor X7R 0805 (2012 Metric) 0.079" L x 0.049" W (2.00mm x 1.25mm)	C106, C301, C305	C0805	C1uF_0805	3	TMK212B7105KG-T
4.7k	RES SMD 1K OHM 5% 1/8W 0805, 4.7k Ohm ±5% 0.125W, 1/8W Surface Mount Resistor Thick Film -55ppm/°C 0805	R325, R402, R403, R408, R409, R410, R411, R412, R413, R414, R416	R0603	R3k_0805, R4.7k_0805	11	CR0805-8W-1022T, RC0805FR-074K7L
8.2k	RES SMD 8.2K OHM 5% 1/8W 0805	R111	R0805	R8.2k_0805	1	RC0805JR-078K2L
10k	Res General Purpose Thick Film 0603 10K Ohm 1% 1/10W ±100ppm/°C Molded SMD Paper T/R, RES SMD 10K OHM 5% 1/10W 0603	R100, R112, R115, R116, R121, R122, R126, R127, R200, R201, R202, R208, R212, R313, R314, R316, R316, R317, R318, R319, R320, R215, R322, R400, R415	R0603	R10k_0603	29	RC0603FR-0710RL, RC1608J103CS
10uF	10µF 10V Ceramic Capacitor X5R 0805 (2012 Metric) 0.079" L x 0.049" W (2.00mm x 1.25mm), 10µF 6.3V Ceramic Capacitor X5R 0805 (2012 Metric) 0.079" L x 0.049" W (2.00mm x 1.25mm)	C105, C112, C114, C116, C120, C401	C0805	C10uF_0805_10V, C10uF_0805	8	UMK212AB7106KG-T, JMK212AB7106KG-HT
11k5	11.5k Ohm ±1% 0.125W, 1/8W Chip Resistor 0805	R139	R0805	R11.5k_0805	1	RC0805FR-0711K5L
15pF	15pF ±1% 50V Ceramic Capacitor C0G, NP0 0805	C200, C201, C303, C307	C0805	C15pF~0805	4	0805C150FK50NXT
22R		R405, R407, R408	R0805	R22R~0805	3	
32.768k	32.768kHz ±20ppm Crystal 12.5pF 50 kOhm -40°C ~ 85°C Surface Mount Cylindrical Can, Metal	Y200	TF-206	Crystal_EPSON_32.768_6_0pF~20ppm	1	EC3~ 327-12.5-12K1-A-C
33pF	33pF ±1% 50V Ceramic Capacitor C0G, NP0 0805 (2012 Metric)	C414, C415	C0805	C33pF~0805	4	CC0805FR-NP0BBN330
47k	47kOhms ±5% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) Moisture Resistant Thick Film	R125	R0805	R47k_0805	1	RC0805JR-074K7L
47mH	FIXED IND 47mH	L405	L0603	L4.7mH_0603	1	
100k	RES SMD 100K OHM 5% 1/10W 0603	R105, R116, R128, R200, R226	R0805	R100k_0603	6	[NoParam], RC0603JR-07100KL
100R		R135	R0805	R100R	1	ERJ-U08G100RV
100uF	Ceramic Capacitor, Multilayer, Ceramic, X~3V, 20% ~Tol, 20% ~Tol, X5R, 15% TC, 100uF, Surface Mount, 1206, 100µF 16V Ceramic Capacitor X5R 3216 (3225 Metric) 0.126" L x 0.098" W (3.20mm x 2.50mm)	C130, C133, C111, C131, C122, C407, C408	C1210	C100uF_1210_10V	7	GMC31X5R107M6 03NT, UMJ325B7107MM-T
102	1000pF ±10% 100V Ceramic Capacitor X7R	C117	C0805	C1nF~0805	1	UMJ212KB7102KD-HT
120R	120 Ohms ±1% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) Moisture Resistant Thick Film	R103, R310	R0805	R120R	2	RC0805FR-07120RL
200k	SMD 200K OHM 1% 1/8W 0805	R131, R102, R110, R118, R124, R125, R816	R0805	R200k_0805	7	RT0805FR0720K0L
470k	SMD 470K OHM 1% 1/8W 0805	R139	R0805	R200k_0805	1	
1000uF_16V	CAP ALUM 330UF 20% 16V RADIAL	C130, C105	C_10*17mm	C_330uF~16V	2	UVR1C102MHD1TD
Battery Management 2.1A, 1- or 2-Cell Li-Ion Battery Charg	Battery Charger Li-Ion/Li-Poly 2.1A 5.95V to 18V 16-Pin QFN T/R	S101	MP2615CGQ-P	MP2615CGQ-P	1	MP2615CGQ-P
Battery Protection IC	One Cell Lithium-Ion/Polymer Battery Protection IC	S100	DW01	DW01	1	
C100pF	100pF ±1% 50V Ceramic Capacitor C0G, NP0 0603 [1608 Metric]	C402	C0603	C100nF 0603 1%	1	GCM188C1H101FA16D
C470uF-16V	470µF 16V Aluminum Electrolytic Capacitors Radial, Can 2000 Hrs @ 85°C	C124	CP_6*12mm	C470uF~16V	1	UVR1C471MFD1TD
CON_1		H100, H101, H102, H103, H104, H105, H106, H303, H304, H305, H306, H307, H308, H309, H310	Connector_1	Con 1	15	
Cong Jac 1u		D300	Magnetic switch	Magnetic switch	1	MH2O2B3
Diode Schottky 20V 2A	VR=20V IF=500mA 170pF P=410mW	D100	DO5FL36BW30-JN	Diode Schottky 20V 2A	1	B5S20LW-7-F
DN25QH8VBA-104HSF	IC FLASH 64MBIT	U101	S25FL128S	S25FL128K	1	
Ferrite bead 0805	FERRITE BEAD 25 OHM 0805 1LN	L109	L0805	Ferrite bead 0805	1	CBG3F1S2HZ	
Fiducial		FM100	Fiducial marker	Fiducial	3	
HT	hole_3mm	*101, *103, *102, *104	Hole_3mm	Hole_4mm	4	
Header 2H_j	Header, 2-Pin, Right Angle	H107	header_2P_x52.54	Header 2H_j	1	
Header 2H_j	Header, 2-Pin, Right Angle	H108, H109, H200, H300, H301, H302	HDR1X2	Header 2H_j	6	
Header 3P 1X2.54		P205	HDR1X3	Header 3P 1X2.54	1	
Header 4	Header, 4-Pin	P203	HDR1X4	Header 4	1	
IRLML6246TRPBF	Mosfet , 20V , 4.1A , 45 Mohm , 3.5 Nc Qg , 2.5V Drive Capable, SOT-23	F102, F103, F108	SOT23A	IRLML6246TRPBF	3	IRLML6246TRPBF
IRML6402PbF	MOSFET P-CH 12V 4.3A SOT-23	F104, F105, F106, F107, F110, F111, F200, F300, F301, F400	SOT23A	IRML6402PbF	10	IRLML6402TRPBF
Kbe Anteen Receptacle U.FL		F401, F402	Kbe Anteen Receptacle U.FL	Kbe Anteen Receptacle U.FL	2	
L_4.7uH_2.6A	4.7uH ±20% 0.04Ω 39MHz	L201	Inductor-6x6mm	L_4.7uH	1	NR6028T4R7M
L_6.8uH_2.6A	6.8uH 2.6A	L200	Inductor-6x6mm	L_4.7uH	1	AAPD0630CD6R8MT
L_BLUE	Blue 470nm LED indication - Discrete 0.06V 0805	D400, D401, D402	LED0805_Blue	LED_0805_bl	3	
led_RGB_5mm		D300	LED_RGB_plh.5	LED_RGB	1	
logo		Logo Sep vinil100	Logo Tag-yan	logo	1	
MCU	Ultra-low-power with FPU ARM Cortex-M4 MCU 80 MHz with 256 Kbytes Flash, LCD, USB	U200	LQFP64	STM32L433RCT6	1	STM32L433RCT6
MFF2		D400	SSM_MFF2	SSM	1	
MMBT3904LT1G	TRANS NPN 40V 0.2A SOT23	T300, T101, T102, T103, T400, T401, T402, T403, T404, T405	SOT23A	NPN. MMBT3904LT1G	10	MMBT3904LT1G
MPDUAF_4G		U400	SF2060J	AC2060J	1	
MOSFET N-2CH		F101	FSB50GA	FSB50GA	1	
MOSFET P-CH 30V 3.6A	Mosfet P-ch 30V SOT23 / Trans Mosfet P-ch 30V 3.6A 3-PIN SOT-23 T/r	F100	SOT23A	DMP3099L-7	1	DMP3099L-7
NTC_10k	Ntc Thermistor, 10Kohm, Radial Leaded Rohs Compliant: Yes	R114	HDR1X2	R3k_0805	1	NXRT15XH103FA1B030
R10R	RES SMD 10 OHM 10% 1/8W 0805	R404	R0805	R10R_0805	1	SR0805KR-0710RL
SN6504V/D485E	IC 5X3X 6540S/RS422 6-SOIC	U300	Sop-8	MAX485 ENBSAVD7ZDR	1	SN6504V/D485E
SS34	40V PURP 1AV 1A SMA	D103	DO-214AC	Diode_SCHOKY 1AV 1A 35A	1	SS34-HF
TPS7A3333PDBVR	IC REG LINEAR 3.3V 200MA SOT23-5	U105	SOT23-5_L	REG113NA-3.3/3K	1	
TPS61088TDGKR		U102	TPS61088TDGKR	TPS61088TDGKR	1	TPS61088TDGKR
Value	220µF 16V Aluminum Capacitors Radial, Can - SMD 2000 Hrs @ 85°C	C412, C413	CAP_ELEC_5x5.7	C220u~16V	2	EEE-1CA221XP